

Algebra I - Introduction to Quadratic Functions Worksheet

Graph the following quadratic functions. Find the vertex and two other points by using an x/y chart.

1. $y = x^2 + 4x + 3$ $a=1$ $b=4$ $c=3$

Vertex

$$x = \frac{-b}{2a}$$

$$y = (-2)^2 + 4(-2) + 3$$

$$x = -2$$

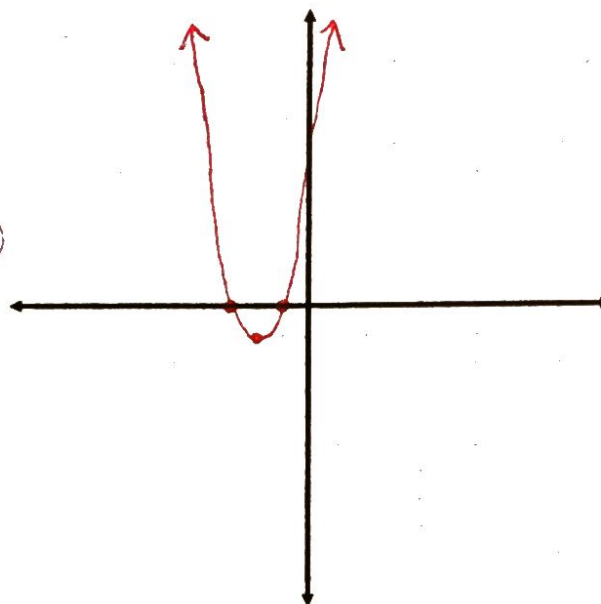
$$y = -1$$

$$(-2, -1)$$

x	y
-3	0
-1	0

$$(-3, 0)$$

$$(-1, 0)$$



2. $y = x^2$ $a=1$ $b=0$ $c=0$

Vertex

$$x = \frac{-b}{2a}$$

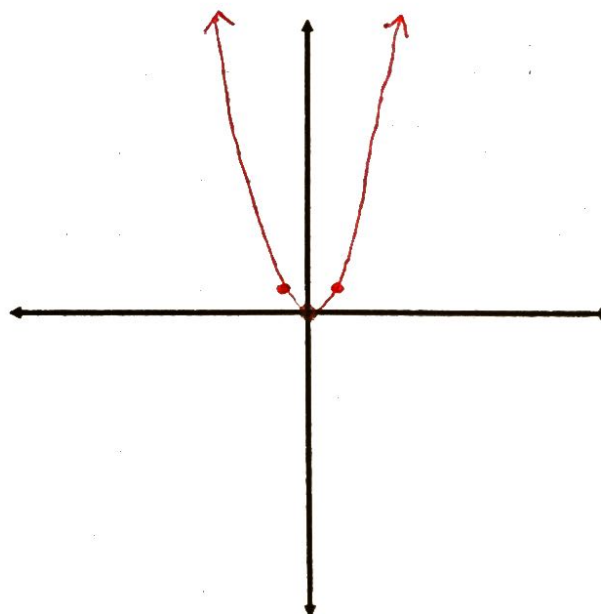
$$y = 0^2$$

$$x = 0$$

$$y = 0$$

$$(0, 0)$$

x	y
1	1
-1	1



Graph the following quadratic functions. Find the vertex and two other points by using an x/y chart.

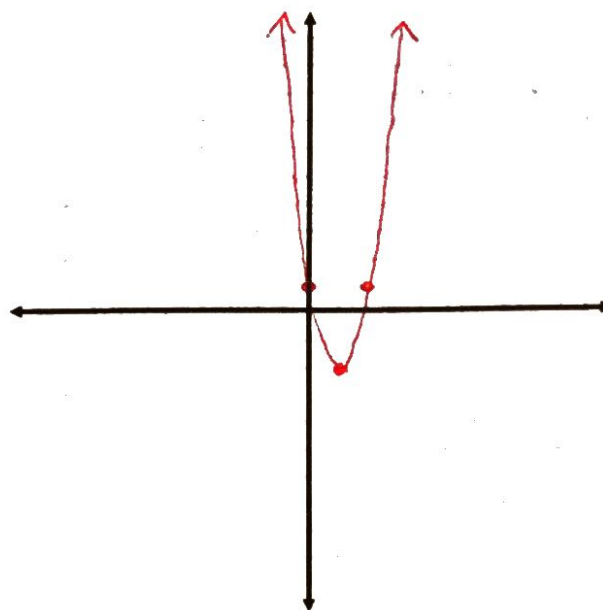
3. $y = 3x^2 - 6x + 1$ $a=3$ $b=-6$ $c=1$

Vertex

$$x = \frac{b}{2a} \quad y = 3(1)^2 - 6(1) + 1$$

$$x = 1 \quad y = -2 \quad (1, -2)$$

x	y
0	1
2	1



4. $y = -2x^2 + 8x$ $a=-2$ $b=8$ $c=0$

Vertex

$$x = \frac{-b}{2a} \quad y = -2(2)^2 + 8(2)$$

$$x = 2 \quad y = 8 \quad (2, 8)$$

x	y
3	6
1	6

